The actual normalizer determines both upper class groups at fifty-three

E.121. Full upper S-class groups, trace maps and unit defects [Proved]

Retain the compatible roots and prime labels of E.119:

$$\begin{gathered}\zeta=\zeta_3,\quad \xi=\zeta_9,\quad \alpha=53^{1/3},\quad \beta_1=\alpha\xi,\quad \beta_2^3=\beta_1,\\ L_i=\mathbb{Q}(\beta_i),\quad B_i=L_i(\xi),\quad F=\mathbb{Q}(\xi),\quad A_D=\mathrm{Cl}_S(D)\{3\},\quad S=\{v : v\mid 3\cdot 53\}.\end{gathered}\tag{E121.1}$$

The roots of positive rational numbers are positive, and the roots of unity are the fixed compatible ones. On B_2 put $g(\beta_2)=\xi\beta_2$, $g(\xi)=\xi$, $h(\beta_2)=\beta_2$, $h(\xi)=\xi^4$, and $\sigma=g^3$. The entire upper S-class groups are

$$A_{B_2}=C_9\oplus C_3^3,\qquad \sigma|A_{B_2}=1,\qquad A_{L_2}=C_9^2.\tag{E121.2}$$

Use the E.119 Smith coordinates $x=(x_0,x_1,x_2,x_3)$ on A_{B_2}, with moduli $(9,3,3,3)$, and write e_i for their positive coordinate generators. Set $u=N_{B_2/L_2}(e_0)$. Choose $a\in A_{L_2}$ whose E.119 coinvariant coordinate is $(1,1)\in C_9\oplus C_3$. Then (u,a) is an actual basis of A_{L_2}, both generators having order nine. Its precise ambiguity is $a\mapsto a+3ku$, $k\in\mathbb{Z}/3$.

In these bases the full horizontal maps, with target rows reduced by their displayed moduli, are

$$N_h=\begin{pmatrix}1&0&6&6\\0&3&6&3\end{pmatrix}: A_{B_2}\longrightarrow A_{L_2},\qquad j_h=\begin{pmatrix}0&3\\2&0\\2&0\\0&1\end{pmatrix}: A_{L_2}\longrightarrow A_{B_2}.\tag{E121.3}$$

The lower bases of $A_{B_1}=C_3^2$ and $A_{L_1}=C_3^2$ remain exactly those of E.119. The full vertical norm and extension maps are

$$\begin{aligned}N_B&=\begin{pmatrix}0&0&0&2\\1&0&0&2\end{pmatrix},\quad j_B=\begin{pmatrix}6&3\\0&0\\0&0\\0&0\end{pmatrix},\\ N_L&=\begin{pmatrix}0&2\\2&0\end{pmatrix},\qquad\qquad j_L=\begin{pmatrix}0&6\\6&0\end{pmatrix}.\end{aligned}\tag{E121.4}$$

Thus $\ker j_B=\langle(1,1)\rangle$, $\ker j_L=0$, and $\ker j_h=3A_{L_2}\simeq C_3^2$. Every norm here is the actual ideal norm, and every j is actual ideal extension. The upper positional action and its quotient map are

$$\sigma_L=\begin{pmatrix}1&3\\0&1\end{pmatrix},\qquad p_L=\begin{pmatrix}6&1\\2&1\end{pmatrix}: (\mathbb{Z}/9)^2\longrightarrow \mathbb{Z}/9\oplus\mathbb{Z}/3,\qquad \ker p_L=(\sigma_L-1)A_{L_2}=\langle 3u\rangle.\tag{E121.5}$$

The full normalizer actions on A_{B_2} are

$$G=\begin{pmatrix}1&0&0&0\\2&0&1&2\\2&2&2&2\\0&0&0&1\end{pmatrix},\qquad H=\begin{pmatrix}4&3&6&0\\2&2&2&0\\0&1&0&0\\1&0&0&1\end{pmatrix}.\tag{E121.6}$$

In particular $(g-1)A_{B_2}=\langle(0,1,1,0)\rangle$. The cyclic-nine genus construction over F gives the actual quotient

$$\begin{gathered}(A_{B_2})_g=C_9\oplus C_3^2,\qquad \pi_g=\begin{pmatrix}5&3&6&6\\1&1&2&1\\0&0&0&2\end{pmatrix},\\ (A_{B_2})_h=C_9\oplus C_3,\qquad (A_{B_2})_{g,h}=C_3^2,\qquad \ker\pi_g=\langle(0,1,1,0)\rangle.\end{gathered}\tag{E121.7}$$

These maps compare the groups through their actual class-field and ideal maps. For example the projected horizontal norm $Q_B \to Q_L$ of E.119 has image $3C_9 \oplus C_3$, of index three, even though $Q_{B,h}$ and Q_L both have type $C_9 \oplus C_3$.

There is also a literal normal-closure comparison:

$$\begin{gathered} C = B_2\overline{B_2} = \mathbb{Q}(53^{1/9}, \zeta_{27}), \qquad B_2 \cap \overline{B_2} = B_1, \\ C/B_2 \text{ is cyclic cubic, unramified and S-split,} \qquad \psi_C(x) = x_0 + x_1 + 2x_2 + x_3 \pmod 3. \end{gathered} \tag{E121.8}$$

Here ψ_C is the Artin character of that actual extension, with the positive generator specified in the proof.

Write $U_D = \mathcal{O}_{D,S}^{\times}$, and put a hat with subscript three for completion by multiplication by powers of three. The following finite unit-norm groups and completed norm kernels are determined:

E/D	U_D/NU_E	$(U_D \cap NE^{\times})/NU_E$	$\ker(\widehat{U}_{E,3} \to \widehat{U}_{D,3})$
B_2/B_1	C_3^6	C_3^4	$C_3 \oplus \mathbb{Z}_3^{18}$
L_2/L_1	C_3	C_3	$C_3 \oplus \mathbb{Z}_3^6$
B_2/L_2	C_3	C_3	$C_3 \oplus \mathbb{Z}_3^{22}$.

(E121.9)

The finite quotients and their actual defining norm subgroups are unchanged by completion. Formula (E121.20) gives their element-to-ideal comparison and inverse; no uncomputed upper unit basis is assigned a numerical matrix. This settles the full next lower positional transition at 53. It does not assign the same groups to subsequent levels. The comparison to the global H-section norm system remains E.46's literal root-evaluation and root-cubing Kummer–Shapiro map, and the further intrinsic maps remain the quotients of E.38. The manuscript anchor is §51.16, scans172–178, folios733v–736v, equations(371)–(379), and Part E(c),(d).

Proof. Fields and the actual genus quotient over F. The polynomial $X^9 - 53\zeta$ is Eisenstein at each prime over 53 of F, so $[B_2 : F] = 9$ and g is its specified generator. The formulas for h preserve $\beta_2^9 = 53\zeta$ and give $hgh^{-1} = g^4$, $h^3 = 1$. Their restrictions to B_1 are $g(\alpha) = \zeta\alpha$ and $h(\alpha) = \zeta^{-1}\alpha$, $h(\xi) = \xi^4$.

The equality $v_3(53 + 1) = 3$ places 53 in the ninth powers of $\mathbb{Q}_3^{\times}$: on $1 + 3\mathbb{Z}_3$ this follows from the convergent logarithm and multiplication by nine, and the sign has its compatible odd root. Thus the completion of B_2/F at three is $\mathbb{Q}_3(\zeta_{27})/\mathbb{Q}_3(\xi)$, of degree three. At the three primes over 53 the local degree is nine. There is no other finite ramification.

Let Γ_F be the Galois group over F of the maximal subfield of $H_{B_2}^S$ abelian over F. The local-to-global Artin quotient of E119.23, now over F, is generated in order (t_1, q_3, t_2, t_3) by local groups of orders $(9, 3, 9, 9)$. The local maps are induced by the proved cup-product norm square of [1, III,Theorem3.1]; their global product is [1, V,Proposition5.2,Theorems5.3 and5.5]. Existence of the class field and its prescribed splitting comes from those exact idelic subgroups.

For $\lambda = 1 - \xi$, the absolute norm of λ is three. The arithmetic local reciprocity characters, with the same global ξ at all three tame places, are

$$\begin{aligned} \chi_q(f) &= -\frac{N_{F_q/\mathbb{Q}_3}(f)/3^{v_q(f)} - 1}{9} \pmod 3, \\ \chi_{t_i}(f) &= -\log_{\bar{\xi}_i}\left(\overline{f/(53\zeta)^{v_{t_i}(f)}}\right)^{312} \pmod 9. \end{aligned} \tag{E121.10}$$

The expression in the first numerator is in $1 + 9\mathbb{Z}_3$. The norm uniformizers are λ and 53ζ respectively; the latter is the norm of β_2 in the cyclic-nine extension. Arithmetic reciprocity acts on cyclotomic roots by the inverse unit norm, which fixes the first sign. The tame symbol

gives the second. Both are evaluated using actual residue fields, not an unspecified relabeling of their roots of unity.

The complete three-completed S-unit basis of the class-number-one field F, certified in E.119, is

$$\xi,\ 1+\xi,\ 1+\xi^2,\ \xi^4-1,\ -4+\xi^4+\xi^5,\ 3+2\xi+3\xi^2+3\xi^3-\xi^4+\xi^5,\ -4-\xi+\xi^2-\xi^4.$$

$$C_F=\begin{pmatrix}0&0&0&0&0&0&0\\3&6&3&6&0&0&0\\3&6&3&6&0&0&0\\3&6&3&6&0&0&0\end{pmatrix}\quad\text{in row order }(q_3,t_1,t_2,t_3).\tag{E121.11}$$

Their S-valuation columns are zero for the first three and the four standard basis columns for the last four. Thus all global relations have been included. In the generator order (t_1,q_3,t_2,t_3) the relation HNF and restriction to $\mathrm{Gal}(B_2/F)$ are

$$H_F=\begin{pmatrix}9&0&0&3\\0&3&0&0\\0&0&9&3\\0&0&0&3\end{pmatrix},\qquad \rho_F=(1,3,1,1).\tag{E121.12}$$

Their full Smith maps give $\Gamma_F=C_9^2\oplus C_3^2$ and $\ker\rho_F=C_9\oplus C_3^2$. For example the latter has the following explicit representatives and inverse coordinates:

$$R_F=\begin{pmatrix}12&-3&-11\\0&1&0\\-2&0&1\\-1&0&1\end{pmatrix},\qquad P_F=\begin{pmatrix}0&0&-1&1\\0&1&0&0\\0&0&-1&2\end{pmatrix}.\tag{E121.13}$$

One checks $P_FR_F=I_3$, $\rho_FR_F\equiv 0 \bmod 9$, and the relations of moduli $(9,3,3)$ in both directions. At a prime over 53, the decomposition group in $H^S_{B_2}/F$ has order nine, because $H^S_{B_2}/B_2$ is split there. It gives an actual section of the cyclic restriction. If $A=A_{B_2}$, the commutator subgroup of $A\rtimes\langle g\rangle$ is $(g-1)A$. Consequently $\ker\rho_F=A_g$ with its actual quotient map.

Exact normalizer and horizontal maps before any upper class deduction. Let Γ_B, H_B, ρ_B, and the generators $(E_1,\dots,E_6,G_1,G_2,G_3)$ be precisely those of E119.21–26. The wild generators have the analytic prime order (q_0,q_1,q_2); the original prime-list order is transported by its retained exact permutation. The actions on the six local generators are the permutations $g:(3,1,2,4,5,6)$ and $h:(2,3,1,6,4,5)$. Both normalizers commute with σ, so the coefficient on each transported local generator is one.

For every G_i the program computes the literal ideal identity $f(G_i)=(c_{f,i})\prod_j G_j^{C_{f,ji}}$ and verifies its full prime factorization. In the class basis and local-character basis just fixed, the matrices are

$$C_g=\begin{pmatrix}1&6&0\\1&1&0\\0&0&1\end{pmatrix},\quad C_h=\begin{pmatrix}4&3&3\\2&1&0\\2&0&1\end{pmatrix},$$
$$X_g=\begin{pmatrix}0&1&0\\0&0&2\\1&2&1\\0&1&1\\0&0&1\\0&2&1\end{pmatrix},\quad X_h=\begin{pmatrix}2&0&2\\0&1&2\\2&2&2\\2&1&2\\1&2&0\\1&0&1\end{pmatrix},\qquad A_f=\begin{pmatrix}P_f&-X_f\\0&C_f\end{pmatrix}.\tag{E121.14}$$

The minus sign is the product formula for the principal idele $c_{f,i}$. All its factors are literal algebraic elements, and every rational prime in the factorization is proved prime. The checks $H_B^{-1}A_fH_B \in M_9(\mathbb{Z})$ and $\rho_B A_f = \rho_B \bmod 3$ prove descent and preservation of the actual restriction kernel.

The actual ideal norm from B_1 to L_1 and ideal extension back give the following maps on their idelic genus generators:

$$\mathcal{N} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 & -1 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix},$$
$$\mathcal{J} = \begin{pmatrix} 1 & 0 & -2 & -1 \\ 1 & 0 & 0 & -2 \\ 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}. \tag{E121.15}$$

For norms, every factor prime is contracted by evaluating the actual embedded lower-field uniformizer at it; its exponent is multiplied by the actual relative residue degree. For extension, the embedded lower ideal generators generate the upper ideal, whose full norm and principal correction are verified. Hence these matrices are arithmetic maps. Their unreduced identities are $\mathcal{N}\mathcal{J} = 3$ modulo the lower relation lattice and $\mathcal{J}\mathcal{N} = 1 + A_h + A_h^2$ modulo the upper lattice.

The corresponding norm to the cyclic-nine presentation over F is

$$\mathcal{P} = \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & -2 & -3 & -4 \\ 1 & 1 & 1 & 0 & 0 & 0 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 3 & 0 & -1 & -3 & -7 \\ 0 & 0 & 0 & 0 & 0 & 3 & -3 & -3 & -1 \end{pmatrix}. \tag{E121.16}$$

At three the local norm is the identity on each split completion; at 53 it includes $\langle g^3 \rangle$ in $\langle g \rangle$, giving the coefficient three. The principal corrections of the three actual ideal norms give the last three columns. Directly $H_F^{-1}\mathcal{P}H_B$ is integral and $\rho_F\mathcal{P} = 3\rho_B \bmod 9$. Passing through the complete Smith maps gives exactly (E121.6)–(E121.7), the projected norm $\bar{N}(x) = (6x_0 + 3x_1 + 6x_2 + 3x_3, 2x_0)$, and projected extension $\bar{j}(r,s) = (3r, 2r+s, 2r+s, r)$.

All these calculations precede the conclusion about full upper class groups. The ambient Smith forward formula is $U_B K_B^{-1}$, integral on the actual lattice $K_B\mathbb{Z}^9 = \ker(\rho_B : \mathbb{Z}^9 \to C_3)$. The retained certificate encodes its ambient rational entries exactly and verifies both inverse compositions. No division in an ambient integer lattice is suppressed.

The finite-module argument closes the entire B_2 kernel. On the actual finite three-group $A = A_{B_2}$ put $T = g - 1$, $s = \sigma - 1$, and $M = sA$. The computed action on A/M has $(G - I)^2 = 0$, hence $T^2A \subset M$. The lower g-action on A_{B_1} is trivial: E.119 proves this using its actual quadratic norm/extension and $cgc = g^{-1}$, and (E121.14) independently checks it on the generating genus columns. Functoriality of actual ideal norm and extension therefore gives

$$\begin{gathered} T(1 + \sigma + \sigma^2)A = Tj_B N_B A = 0, \qquad 3TA = -T(3+s)M, \\ M = (T^3 + 3T^2 + 3T)A \subset TM + 3M + T(3+s)M \subset (3, T)M. \end{gathered} \tag{E121.17}$$

Here $j_B N_B = 1+\sigma+\sigma^2$ is the literal product of the conjugate ideals, not an abstract module identification. Choose e with $3^e A = 0$. Since $g^9 = 1$, $T^9 \in 3\mathbb{Z}_3[g]$. Every monomial of total degree $9e$ in $(3,T)$ is consequently divisible by $3^{a+\lfloor b/9 \rfloor}$ with $a+b = 9e$, and that exponent is at least e. Thus $(3,T)^{9e}A = 0$. Iterating (E121.17) gives $M = 0$. This proves the full A_{B_2} assertion, with all E.119 quotient maps now the full class maps. In particular j_B has exactly the diagonal kernel stated above. The matrices G, H are now the full actual normalizer actions.

The positive connecting class determines A_{L_2}. The cubic B_2/L_2 is unramified and split at S. Indeed it adjoins ξ; at three L_2 already has its $\mathbb{Q}_3(\zeta_{27})$ completion, and at 53 the residue field contains μ_9. Thus $B_2 \subset H^S_{L_2} \subset H^S_{B_2}$. The maximal subfield of $H^S_{B_2}$ abelian over L_2 is exactly $H^S_{L_2}$, by these actual local conditions. For a cyclic quotient generated by a lift of h, the commutator subgroup is $(h-1)A_{B_2}$. The ideal norm/Frobenius square of [1, V,Proposition3.3] therefore gives the exact sequence

$$0 \longrightarrow (A_{B_2})_h \xrightarrow{N_h} A_{L_2} \xrightarrow{\mathrm{Art}_{B_2/L_2}} C_3 \longrightarrow 0, \qquad |A_{L_2}| = 3 \cdot 27 = 81. \tag{E121.18}$$

On the E.119 quotient Q_L its last map is the first coordinate modulo three. This is also the restriction of the actual B_1/L_1 character, whose lower idele row is $(0,0,2,0)$ in E.119.

For the connecting class choose the actual inertia lift σ_0 at the first 53-prime in $H^S_{B_2}/B_1$. It has order three and its image in Γ_B is E_4. Let h_0 lift h while fixing L_2. With commutator $[x,y] = xyx^{-1}y^{-1}$, the image of $[\sigma_0, h_0]$ in Γ_B is $E_4 - A_h E_4 = E_4 - E_6$. Its exact Smith coordinate is $(3,0,0,0)$. Changing either lift changes the class by $(h-1)A_{B_2}$ or $(\sigma-1)A_{B_2}$, which are precisely the subgroups being quotiented. Thus the homology connecting map sends the positive $\sigma \otimes h$ to $[3e_0]$. This element is nonzero in $(A_{B_2})_h$.

Write $u = N_h(e_0)$. The relations from $(H-I)A_{B_2}$ are $e_2 = -3e_0 - e_1$ and $e_3 = -3e_0 - 2e_1$, so u has order nine. Since $\sigma = 1$ on A_{B_2}, the image in (E121.18) is fixed by σ. The quotient Q_L has order 27, so $(\sigma-1)A_{L_2}$ has order three. The connecting calculation gives exactly $(\sigma-1)A_{L_2} = \langle 3u \rangle$. Its square is zero and it is killed by three. Consequently the actual vertical trace is $1+\sigma+\sigma^2 = 3$ on A_{L_2}. As A_{L_1} has exponent three, $j_L N_L = 3$ implies $9A_{L_2} = 0$. Furthermore $(\sigma-1)A_{L_2} \subset 3A_{L_2}$, so $A_{L_2}/3A_{L_2} \simeq Q_L/3Q_L$ has dimension two. These order, exponent and generator constraints prove $A_{L_2} = C_9^2$.

Choose a with $p_L(a) = (1,1)$. Its Artin value is one, so $\sigma a = a + 3u$. The two classes u, a are independent modulo three: u is primitive of order nine and has Artin value zero, whereas a has value one. They therefore give the claimed basis. Put $v = N_h(e_1)$. From the projected maps, $v = 3ku + 3a$ for some k mod 3. The actual identity $N_h j_h(a) = 3a$, with $j_h(a) = (3,0,0,1)$, now gives $-6ku + 3a = 3a$ and hence $k = 0$. Thus the two exact power relations are

$$3a = N_h(e_1), \qquad \sigma a - a = 3u. \tag{E121.19}$$

They prove all full horizontal matrices in (E121.3). The lower norm square gives $N_L(u) = (0,2)$. Applying the actual projected norm of E.119 to $p_L(a) = (1,1)$ gives $N_L(a) = (2,0)$. It is onto; alternatively, onto-ness follows directly from E.119's onto quotient norm. The trace identity then forces $j_L = 3N_L^{-1}$ in the displayed bases, yielding (E121.4). Its kernel is zero. All norm/extension squares and both compositions are verified on the full finite groups. The ambiguity $a \mapsto a + 3ku$ gives the literal coordinate shear $(r,s) \mapsto (r+3ks, s)$; it leaves every displayed map unchanged with its stated source and target moduli.

The normal closure and its Artin character. Put $\alpha_2 = 53^{1/9}$ and $\eta = \zeta_{27}$, so $\beta_2 = \alpha_2\eta$. The literal inverse formulas $\alpha_2 = (\beta_2\overline{\beta_2})^5/53$ and $\eta = \beta_2/\alpha_2$ identify the compositum in (E121.8). Its degree is 162, by the 53-valuation and the cyclotomic degree. On C, use $a_0(\alpha_2) = \xi\alpha_2$, $a_0(\eta) = \eta$, and $b_0(\alpha_2) = \alpha_2$, $b_0(\eta) = \eta^4$. Both have order nine and $b_0 a_0 b_0^{-1} = a_0^4$. The subgroup fixing B_2 is generated by $a_0^{-3}b_0^3$, and the conjugate subgroup is generated by $a_0^3 b_0^3$. Their intersection is trivial and the group they generate fixes B_1. These literal subgroup maps prove

the intersection assertion. They also show exactly why complex conjugation does not act on B_2 itself.

The completions of C and B_2 agree at each place of S: at three they are the stated cyclotomic completion, and at 53 the residue field contains μ_{27} because $v_3(53^2-1)=3$. The discriminants have no other finite ramification, so C/B_2 is unramified and S-split. In the presentation (E121.12) its map to $\mathrm{Gal}(C/F)=C_9\oplus C_3$ is $(t_1,q_3,t_2,t_3)\mapsto(t_1+t_2+t_3,q_3)$. Restriction to B_2 is $(A,B)\mapsto A+3B \bmod 9$. On its kernel the positive generator is $a_0^{-3}b_0^3$ and the character is B. Composing this actual map with (E121.16) and the E.119 Smith representatives gives exactly ψ_C in (E121.8).

Actual unit norm subgroups and completion. For any of the three cyclic extensions E/D here, write τ for its positive generator. Outside S the extension is unramified. For $u=Nx\in U_D$, choose an outside-S divisor D_x with $\mathrm{div}_{\notin S}(x)=(\tau-1)D_x$. Such a divisor exists orbit by orbit from the zero norm of the divisor of x. The literal map is

$$\frac{U_D\cap NE^\times}{NU_E}\xrightarrow{\nu}\frac{A_E^\tau}{jA_D},\qquad Nx\longmapsto \mathrm{pr}_3[D_x] \bmod jA_D. \tag{E121.20}$$

Here pr_3 is the canonical projection of the finite S-class group to its three-primary subgroup. On its prime-to-three part, an invariant class is extended from the base because $jN=1+\tau+\tau^2=3$ there and multiplication by three is invertible. Hence the omitted invariant quotient is zero. Changing x by a norm-one element uses Hilbert90 and changes D_x by a principal divisor and an extended base divisor. Conversely an invariant ideal class supplies a representative D for which $(\tau-1)D$ is principal; a generator x has $Nx\in U_D$. These constructions are inverse. This is the actual ideal/Hilbert90 map of [2, Proposition1 and its full ideal proof], with the splitting convention at S retained.

In the three cases its right side is respectively $A_{B_2}/\langle 3e_0\rangle=C_3^4$, $A_{L_2}^\sigma/3A_{L_2}=C_3$, and $A_{B_2}^h/j_hA_{L_2}=C_3$. Explicit quotient coordinates are $(x_0 \bmod 3,x_1,x_2,x_3)$ in the first, $r \bmod 3$ on a fixed class $(r,3s)$ in the second, and $r-s \bmod 3$ on a fixed class $(3r,t,t,s)$ in the third. E.119 evaluates the first field-norm index as nine and the second as one. In the unramified S-split horizontal extension every S-unit is a local norm, hence a field norm by the complete cyclic Hasse norm theorem [1, VIII,Theorem3.1]. Multiplying these exact indices proves (E121.9). Each finite quotient is killed by three because the included base unit has norm its cube. Its order therefore gives the stated elementary abelian type.

For comparison, the capkernel/unit-Herbrand maps of E119.16 give the same indices: the S-unit Herbrand quotients are 3^5, 3, and $1/3$, and the actual capitulation orders are 3, 1, and 9. The precise lattice quotient is [1, VII,Proposition3.1]. These are compatible checks of the same norm subgroup.

The completed unit ranks are $14,32,4,10$ for B_1,B_2,L_1,L_2 respectively. Their three-primary roots of unity are μ_9,μ_9,μ_3,μ_3: $\mu_{27}\not\subset B_2$ by the actual cubic compositum, and $\mu_9\not\subset L_2$ because $[B_2:L_2]=3$. The vertical root norms are cubing. Horizontally $N_h(\xi)=\xi^{1+4+7}=\zeta$. Thus every completed kernel has exactly the displayed literal μ_3 torsion and the rank difference in (E121.9). Finite-rank unit groups complete exactly, and all norm images have the proved finite index, so the same finite quotients and actual images persist under completion.

The five GP stages use only the immutable, already certified E.119 lower-field objects. They prove each ideal relation, local character and prime contraction; the separate exact Python certificate verifies every Smith inverse, all finite class maps, the positive connecting class and the complete field subgroup comparison. No upper BNF certification is used. The retained earlier interrupted attempts supply no additional class-group assertion. □

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